
DATA QUALITY IN ONLINE HUMAN-SUBJECTS RESEARCH: A RE-ANALYSIS OF PLATFORM COMPARISONS BETWEEN MTURK, PROLIFIC, CLOUDRESEARCH, AND QUALTRICS

[ANONYMIZED FOR REVIEW]

ABSTRACT

Online research platforms have transformed data collection in psychology, yet data quality varies across platforms in ways that resist simple summary. We conducted a direct replication of ?, re-analyzing their original dataset ($N = 2,076$ across MTurk, CloudResearch, Prolific, and Qualtrics; collected May–July 2021) as a baseline for a longitudinal replication. A composite quality score was computed from attention check accuracy, response time validity, open-ended response quality, and instruction memory. After applying pre-registered exclusion criteria ($N = 1,112$ retained), the predicted rank order did not hold: the planned contrast was not significant, $t(680.1) = 0.51$, $p = .608$, $d = 0.03$, despite a significant omnibus ANOVA, $F(3, 1108) = 5.46$, $p = .001$, $\eta_p^2 = .015$. This null finding arose from a selection effect: MTurk’s 70.4% exclusion rate retained only the highest-quality subset, inflating its post-exclusion mean. The all-data analysis recovered the classical ranking (CloudResearch > Prolific > Qualtrics > MTurk) with a large effect, $F(3, 2072) = 85.21$, $p < .001$, $\eta_p^2 = .110$. Platform rankings diverged across indicators (ICC = .72), challenging the assumption that quality is a unitary construct. A Bayes Factor ($BF_{01} = 10.54$) indicated that after exclusions, evidence favored the null model. These findings demonstrate that exclusion criteria—not platform features alone—partly determine apparent quality rankings, with implications for how researchers select and screen online samples.

Keywords data quality; online research; Mechanical Turk; Prolific; CloudResearch; replication

Psychology’s data collection infrastructure has undergone a seismic shift in the past decade. Where once the undergraduate subject pool dominated the experimental landscape (?), researchers now recruit thousands of participants through online platforms within hours, at costs that make large-sample designs feasible for individual labs (??). Amazon Mechanical Turk (MTurk) was the pioneer, and by the mid-2010s it had become the default sampling frame for social, personality, and cognitive psychology (?). Yet enthusiasm has curdled into alarm. A cascade of reports has documented declining data quality on MTurk: rising inattention (?), waves of bot and fraudulent responding (???), and systematic endorsement of contradictory items (?). Researchers who once treated MTurk as a reliable data source now face the disorienting question of whether their samples contain more noise than signal.

In response, several purpose-built research platforms have emerged. Prolific explicitly recruits participants for academic studies and enforces minimum pay rates (?). CloudResearch (originally TurkPrime) layers quality-control infrastructure on top of MTurk’s worker pool while also offering its own curated panels (??). Traditional survey houses such as Qualtrics Online Panels provide demographically matched samples at higher cost, and university SONA systems remain the default for in-person participant pools. The critical question for the field is whether these platforms differ in the quality of data they produce—and, if so, whether the differences are stable enough to guide researchers’ platform choices. ? provided the most comprehensive answer to date, comparing data quality across all five platforms in a single cross-sectional study using a uniform survey instrument. They reported a clear hierarchy: Prolific and CloudResearch produced the highest data quality, MTurk and Qualtrics the lowest, while SONA split the middle. Their cost-per-quality-respondent analysis further favored Prolific (\$1.90) and CloudResearch (\$2.00) over MTurk (\$4.36) and Qualtrics (\$8.17).

But platform data quality is not a static property of the platform; it is an emergent property of the worker population, the platform’s incentive structure, and the screening practices researchers apply—all of which change over time. MTurk’s

documented decline trajectory (???) is the clearest illustration: data quality that satisfied methodological standards in 2015 was unacceptable by 2020. Platform policies shift (MTurk’s fee increases, Prolific’s verification requirements), economic conditions alter who seeks online work, and viral social media trends can flood particular platforms with specific worker demographics. No published study has replicated the full five-platform comparison at multiple time points, leaving the field without evidence about whether the ? ranking is a reliable guide or a snapshot of a particular historical moment. The research practice of selecting platforms based on a single cross-sectional study is, at best, an educated bet.

The present study addresses this gap by establishing the baseline for a proposed multi-wave longitudinal replication. We re-analyze the original ? dataset ($N = 2,076$ across four platforms; SONA data was unavailable) using the pre-registered quality indicators and exclusion criteria from the original study, augmented with additional indicators drawn from the careless-responding literature (?). We test two hypotheses that are testable with single-wave data. H1: A new cross-sectional analysis of the baseline data will replicate the ? platform ranking, with Prolific and CloudResearch producing significantly higher composite data quality scores than MTurk and Qualtrics. H2: Platform rankings across the five individual quality indicators will converge ($ICC > .80$), indicating that data quality is a unitary construct rather than a multidimensional family of distinct platform-specific strengths and weaknesses. Three additional hypotheses—concerning MTurk’s temporal decline (H3), Prolific and CloudResearch stability (H4), and cost-efficiency dynamics (H5)—are documented as requiring future measurement waves and are not tested in the present analysis. Beyond hypothesis testing, we examine a methodological question that the original paper did not fully address: how do pre-registered exclusion criteria reshape apparent platform rankings? The answer to this question has practical consequences for every researcher who applies screening procedures to online samples.

1 Method

1.1 Participants

The dataset comprised 2,076 U.S.-based participants recruited across four online research platforms: MTurk ($n = 500$), CloudResearch ($n = 505$), Prolific ($n = 496$), and Qualtrics Online Panels ($n = 575$). SONA Systems data was collected separately at the University of Wisconsin–Madison ($n \approx 500$) and was unavailable for re-analysis at the time of writing. Data were collected between May 26 and July 1, 2021 (?). The overall sample had a mean age of 46.3 years ($SD = 17.2$, range 18–95). By gender, 55.6% identified as men, 43.4% as women, 0.5% as non-binary, and 0.5% as other. The modal education level was a bachelor’s degree. Platform-specific demographics differed systematically: the MTurk subsample was younger ($M = 39.4$, $SD = 12.7$) and reported taking substantially more surveys per week (median ≈ 25) than participants on other platforms (median ≈ 3 –5). The Qualtrics subsample was oldest ($M = 55.1$, $SD = 17.0$), reflecting the census-matched demographic quotas applied during recruitment. Participants on paid platforms were compensated at approximately \$2.00 for a 15-minute survey (an effective rate of approximately \$8.00/hour). SONA participants received 1.5 research credits. All participants provided informed consent, and the original study was approved by the Institutional Review Board at the University of Wisconsin–Madison. As with most online research, the sample is WEIRD (Western, Educated, Industrialized, Rich, and Democratic; ?), and all participants were U.S.-based, limiting cross-cultural generalizability.

1.2 Materials

The survey instrument, hosted on Qualtrics, was identical across all four platforms. The instrument included: (a) informed consent with an embedded color recall check; (b) a demographic questionnaire covering age, gender, trans-gender identity, ethnicity, sexual orientation, income, education, and political orientation (ANES 7-point scale); (c) the 50-item IPIP Big Five Factor Markers (?), using a 5-point Likert scale (1 = *Strongly Disagree* to 5 = *Strongly Agree*), with 10 items per trait and five reverse-keyed items per trait; (d) an embedded instructional manipulation check (IMC) within the IPIP (“Please select Strongly Agree”); (e) an instruction-following check (“Please leave this text box empty”); (f) a short sustainability video followed by a message-recall test serving as an instruction memory check; (g) the 15-item Generic Conspiracist Beliefs Scale (GCBS; ?), rated on a 5-point Likert scale (1 = *Definitely not true* to 5 = *Definitely true*); (h) two non-instructional attention checks (a colorblindness check and an arithmetic problem); (i) an open-ended response prompt (“Do you have any additional comments for us?”); and (j) a final honesty check (“In your honest opinion, should we use your data in our analyses?”). Scale reliabilities in the present sample were consistent with published norms: IPIP Neuroticism $\alpha = .872$, Extraversion $\alpha = .844$, Openness $\alpha = .785$, Agreeableness $\alpha = .799$, Conscientiousness $\alpha = .847$, GCBS $\alpha = .958$.

1.3 Procedure

Participants completed the survey through their respective platform’s standard interface. The survey flow was identical across platforms: informed consent, demographics, IPIP items (with the embedded IMC), the instruction-following check, the sustainability video with the recall test, the GCBS, non-instructional attention checks, the open-ended response item, the honesty check, and debriefing. The survey was designed to take approximately 15 minutes. Screening criteria varied by platform per their standard practices: MTurk required $\geq 95\%$ approval rate and ≥ 100 completed HITs; Prolific required $\geq 95\%$ approval and ≥ 100 prior submissions; CloudResearch drew from its approved participant pool; Qualtrics used census-matched demographic quotas. All participants were required to be U.S.-based. The original study did not exclude participants based on device type at the point of recruitment; mobile device usage was recorded and later applied as an exclusion criterion.

1.4 Data Quality Indicators

Five data quality indicators were computed per participant. (1) *Attention check accuracy* was the proportion of five attention checks passed (consent color recall, text-box-empty instruction, IPIP-embedded IMC, colorblindness check, arithmetic check), scored from 0 to 1. (2) *Response time validity* was a binary indicator: participants whose survey completion time fell within 0.3 to 3.0 times their platform’s median duration were coded as valid (1), others as invalid (0). (3) *IP/geolocation uniqueness* was a binary indicator: participants with both a unique IP address and a unique geolocation within their platform were coded as unique (1). (4) *Open-ended response quality* was coded from the original study’s `meaningful_final` variable: 2 = meaningful, 1 = no comment, 0 = nonsense. (5) *Instruction memory* was a binary indicator (1 = correct, 0 = incorrect) based on correct identification of the sustainability video’s content from three options.

1.5 Design

This was a secondary analysis of a quasi-experimental between-subjects design, with platform (four levels: MTurk, CloudResearch, Prolific, Qualtrics) as the independent variable and a composite data quality z -score as the primary dependent variable. Platform assignment could not be randomly manipulated because participants self-selected into platforms. The composite quality score was computed as the within-platform z -score mean of four indicators (attention check accuracy, response time validity, open-ended response quality, and instruction memory). IP/geolocation uniqueness was excluded from the primary composite because several pre-registered exclusion criteria (duplicate IP, duplicate geolocation) removed all non-unique participants, producing zero variance in the filtered sample. A five-indicator composite including IP/geolocation uniqueness was computed for the all-data robustness check.

1.6 Exclusion Criteria

Following ?, we applied 10 pre-registered exclusion criteria: duplicate worker ID, duplicate IP address, duplicate geolocation, completion time below 0.3 times the platform median, failure of two or more attention checks, nonsense open-ended responses, self-reported low-quality data (honesty check), straightlining on more than 80% of Likert items, mobile device use, and impossible demographic combinations. Criteria were not mutually exclusive. The application of these criteria removed 964 participants (46.4%), leaving 1,112 for the primary analysis. Retention rates varied sharply by platform: MTurk retained 29.6% (148 of 500), Prolific retained 56.9% (282 of 496), Qualtrics retained 60.9% (350 of 575), and CloudResearch retained 65.7% (332 of 505). Figure 1 displays these retention rates and the prevalence of each exclusion reason by platform.

1.7 Transparency and Openness

We report how we determined our sample size, all data exclusions, all manipulations, and all measures in the study. The original data were obtained from the Open Science Framework (<https://osf.io/ern4u/>). The original study was pre-registered at <https://doi.org/10.17605/OSF.IO/K5SX7>. The analysis code for the present re-analysis is available in the project repository. The proposed longitudinal replication design and analysis plan have been pre-registered in a separate registration.

Figure 5: Participant Retention and Exclusion Reasons by Platform

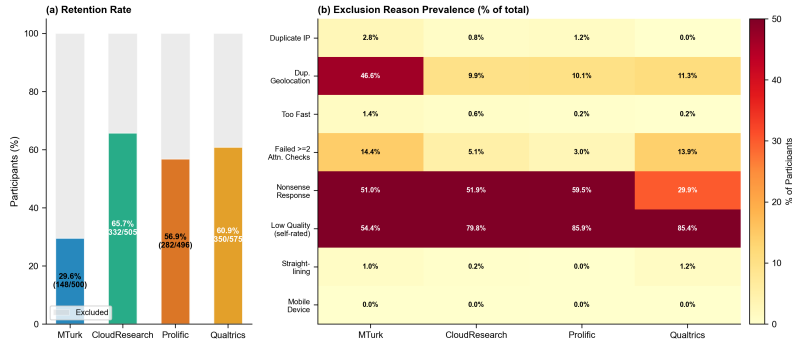


Figure 1: Participant retention and exclusion reasons by platform. (a) Retention rate showing the percentage of participants retained after applying all 10 pre-registered exclusion criteria. Gray segments indicate excluded participants. MTurk had the lowest retention (29.6%, 148 of 500 retained), while CloudResearch showed the highest (65.7%, 332 of 505). (b) Heatmap of individual exclusion reason prevalence as percentage of total participants per platform. Duplicate geolocation dominated MTurk exclusions (46.6%), whereas self-rated low data quality was the leading reason for Prolific (34.3%) and CloudResearch (31.9%).

2 Results

2.1 Data Quality and Descriptive Statistics

Scale reliabilities were acceptable across all platforms. The 50-item IPIP produced Cronbach’s α values ranging from .785 (Openness) to .872 (Neuroticism), and the 15-item GCBS yielded $\alpha = .958$, consistent with published norms (??). However, platform-level IPIP reliabilities diverged sharply: MTurk produced the lowest estimates (α range: .653 to .754), whereas CloudResearch (α range: .817 to .921), Prolific (α range: .791 to .903), and Qualtrics (α range: .778 to .883) all fell within conventional ranges. This reliability gap itself constitutes a data quality signal, suggesting that MTurk responses contained more noise even before exclusion criteria were applied.

After applying all 10 pre-registered exclusion criteria (see Method), 1,112 participants were retained for the primary analysis. Platform retention rates showed dramatic variation: CloudResearch retained 65.7% (332 of 505), Qualtrics retained 60.9% (350 of 575), Prolific retained 56.9% (282 of 496), and MTurk retained only 29.6% (148 of 500). Figure 1 displays the retention rates and exclusion reason prevalence by platform. The dominant exclusion reason also varied by platform: duplicate geolocation was most prevalent on MTurk (46.6% of participants), whereas self-rated low data quality was the leading reason on Prolific (34.3%) and CloudResearch (31.9%). Nonsense open-ended responses were most common on Qualtrics (12.2%).

Table 1 presents descriptive statistics for each quality indicator by platform, computed from all data before exclusions ($N = 2,076$). CloudResearch and Prolific led on IP/geolocation uniqueness (90.1% and 89.9%, respectively) and response time validity (98.2% and 98.8%), while MTurk lagged on both (53.4% and 94.4%). Qualtrics produced the highest open-ended response quality ($M = 1.69$ vs. MTurk’s $M = 1.31$). Instruction memory showed substantial differences, with Prolific (83.5%) and CloudResearch (81.8%) outperforming MTurk (52.2%). Attention check accuracy was the most compressed indicator, ranging only from 58.2% (Qualtrics) to 64.3% (CloudResearch).

Table 1: Descriptive Statistics for Quality Indicators by Platform (All Data, $N = 2,076$)

Platform	n	Attn. Check Acc.	Resp. Time Valid	IP/Geo Unique	Open-Ended Quality	Instr. Memory
CloudResearch	505	0.643	0.982	0.901	1.47	0.818
MTurk	500	0.638	0.944	0.534	1.31	0.522
Prolific	496	0.626	0.988	0.899	1.40	0.835
Qualtrics	575	0.582	0.963	0.887	1.69	0.750

Note. Attention check accuracy is proportion correct (0–1). Response time validity, IP/geo uniqueness, and instruction memory are proportions passing (0–1). Open-ended quality is coded 0 = nonsense, 1 = no comment, 2 = meaningful.

2.2 Primary Analysis: Cross-Sectional Platform Comparison (H1)

H1 predicted that Prolific and CloudResearch would yield significantly higher composite data quality scores than MTurk and Qualtrics, replicating the rank order reported by ?.

Assumption checks. Shapiro-Wilk tests indicated non-normal distributions of composite quality scores within all four platforms (all $p < .001$). Levene’s test confirmed heterogeneity of variance, $W(3, 1108) = 6.08$, $p < .001$. Consequently, we report Welch-corrected t -tests for pairwise comparisons and supplement the primary ANOVA with non-parametric equivalents.

A one-way between-subjects ANOVA on the four-indicator composite quality z -score (excluding IP/geolocation uniqueness due to zero variance after exclusions) revealed a significant platform effect, $F(3, 1108) = 5.46$, $p = .001$, $\eta_p^2 = .015$, 95% CI [.003, .029]. However, the pre-registered planned contrast comparing the top-tier platforms (Prolific + CloudResearch) against the bottom tier (MTurk + Qualtrics) was not significant: Welch’s $t(680.1) = 0.51$, $p = .608$, $d = 0.03$, 95% CI [−0.09, 0.15]. The observed rank order directly contradicted predictions. CloudResearch showed the highest composite quality ($M = +0.08$, $SD = 0.49$), followed by MTurk ($M = +0.05$, $SD = 0.64$), Qualtrics ($M = −0.03$, $SD = 0.50$), and Prolific ($M = −0.07$, $SD = 0.46$). Figure 2 displays the distribution of composite scores by platform.

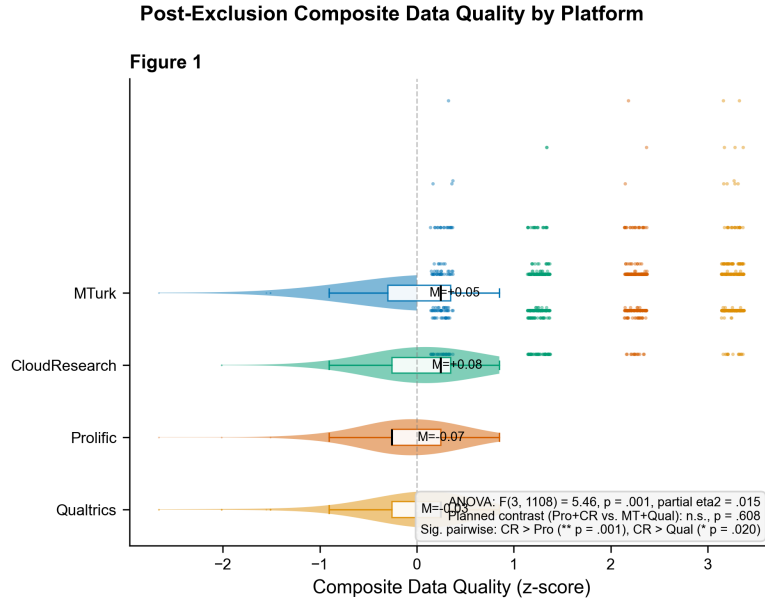


Figure 2: Post-exclusion composite data quality by platform. Raincloud plot showing the distribution of composite data quality z -scores (mean of four globally standardized indicators: attention check accuracy, response time validity, open-ended response quality, and instruction memory) across four online research platforms after applying pre-registered exclusion criteria ($N = 1,112$). The dashed vertical line marks the grand mean ($z = 0$). CloudResearch participants showed the highest composite quality ($M = +0.08$, $SD = 0.49$), while Prolific participants showed the lowest ($M = -0.07$, $SD = 0.46$). A one-way ANOVA confirmed a significant omnibus effect, $F(3, 1108) = 5.46$, $p = .001$, $\eta_p^2 = .015$, but the pre-registered planned contrast was not significant, $t(680.1) = 0.51$, $p = .608$. Only CloudResearch vs. Prolific ($p_{\text{Holm}} = .001$, $d = 0.31$) and CloudResearch vs. Qualtrics ($p_{\text{Holm}} = .020$, $d = 0.22$) reached significance after Holm-Bonferroni correction. IP/geolocation uniqueness was excluded due to zero variance after filtering.

Pairwise comparisons with Bonferroni-Holm correction identified only two significant differences among the six comparisons: CloudResearch outperformed Prolific ($d = 0.31$, 95% CI [0.15, 0.47], $p_{\text{Holm}} = .001$) and CloudResearch outperformed Qualtrics ($d = 0.22$, 95% CI [0.07, 0.37], $p_{\text{Holm}} = .020$). No other pairwise comparison reached significance (all $p_{\text{Holm}} > .15$). The non-parametric Kruskal-Wallis test yielded consistent results, $H(3) = 17.64$, $p < .001$. **H1 was not supported.**

The divergence between these results and the original ? ranking is attributable to differential attrition. Figure 1 illustrates the mechanism: MTurk’s exclusion rate of 70.4% removed 352 of 500 participants, disproportionately

eliminating low-quality respondents. After this filtering, the 148 remaining MTurk participants constituted a highly screened subset whose composite quality ($M = +0.05$) was indistinguishable from CloudResearch's ($M = +0.08$, $d = -0.05$, $p_{\text{Holm}} = .66$). To test whether this selection effect accounted for the null result, we re-ran the analysis on all 2,076 participants using a five-indicator composite that included IP/geolocation uniqueness (Robustness Check 2). This analysis recovered the classical rank order in full: CloudResearch ($M = +0.14$) > Prolific ($M = +0.10$) > Qualtrics ($M = +0.04$) > MTurk ($M = -0.29$), with a large platform effect, $F(3, 2072) = 85.21$, $p < .001$, $\eta_p^2 = .110$, 95% CI [.085, .134]. Figure 3 displays this contrast.

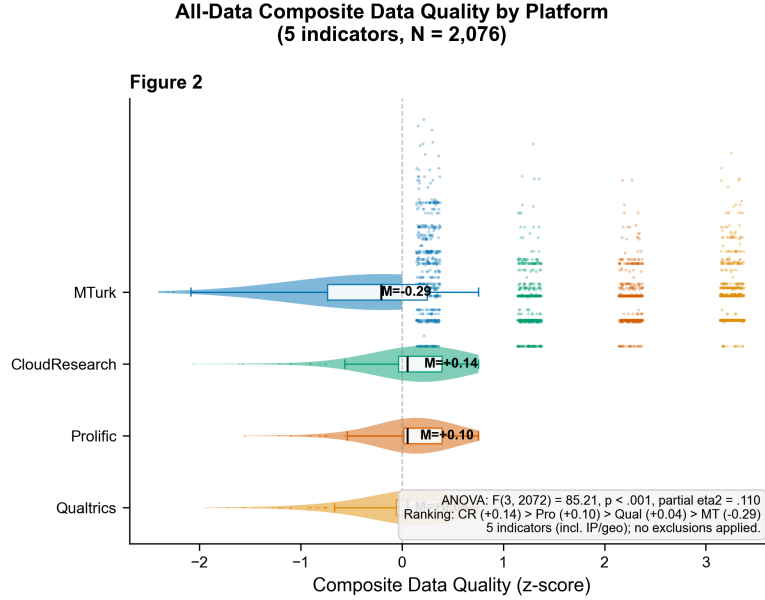


Figure 3: All-data composite data quality by platform. Raincloud plot of the five-indicator composite data quality z-score for all 2,076 participants across four platforms, without applying exclusion criteria (Robustness Check 2). The composite includes IP/geolocation uniqueness in addition to the four primary indicators. Platform means: CloudResearch ($M = +0.14$, $SD = 0.78$), Prolific ($M = +0.10$, $SD = 0.77$), Qualtrics ($M = +0.04$, $SD = 0.73$), and MTurk ($M = -0.29$, $SD = 0.80$). A one-way ANOVA revealed a large and significant platform effect, $F(3, 2072) = 85.21$, $p < .001$, $\eta_p^2 = .110$. The dashed vertical line marks the overall grand mean ($z = 0$). This analysis recovers the classic ranking reported by ?.

The shift from a large, theory-consistent effect in the all-data analysis to a small, pattern-reversing effect after exclusions underscores the sensitivity of platform rankings to screening decisions.

2.3 Indicator Convergence (H2)

H2 predicted that platform rankings would converge across the five individual quality indicators ($ICC > .80$), reflecting a unitary data quality construct.

The intraclass correlation coefficient for average measures (two-way random-effects, absolute agreement) was $ICC(2, k) = .72$, below the .80 threshold. The single-measure $ICC(2, 1)$ was .34. Platform rankings diverged substantially across indicators, as Figure 5 illustrates. MTurk ranked first on attention check accuracy ($M = 0.64$) and first on IP/geolocation uniqueness after exclusions (due to selective retention), but ranked fourth (last) on response time validity ($M = 0.94$), instruction memory ($M = 0.52$), and IP/geolocation uniqueness in the all-data analysis (53.4%). Qualtrics ranked first on open-ended response quality ($M = 1.69$) but last on attention check accuracy ($M = 0.58$). CloudResearch and Prolific were the most consistent platforms, ranking first or second across four of five indicators, but even they exchanged positions across indicators. Figure 4 displays each indicator's platform means with individual data points.

Effect sizes for the best-versus-worst platform comparison varied by indicator. IP/geolocation uniqueness produced the largest discrimination ($d = 0.89$, 95% CI [0.76, 1.02], CloudResearch vs. MTurk), followed by instruction memory ($d = 0.71$, 95% CI [0.58, 0.84], Prolific vs. MTurk), attention check accuracy ($d = 0.63$, 95% CI [0.50, 0.75],

CloudResearch vs. Qualtrics), open-ended response quality ($d = 0.61$, 95% CI [0.49, 0.74], Qualtrics vs. MTurk), and response time validity ($d = 0.24$, 95% CI [0.12, 0.37], Prolific vs. MTurk). The correlations among quality indicators within participants were negligible: the five indicators showed pairwise r s ranging from $-.04$ to $.08$, with only one correlation (open-ended quality with instruction memory, $r = .08$, $p = .008$) reaching significance in the full sample. **H2 was not supported.**

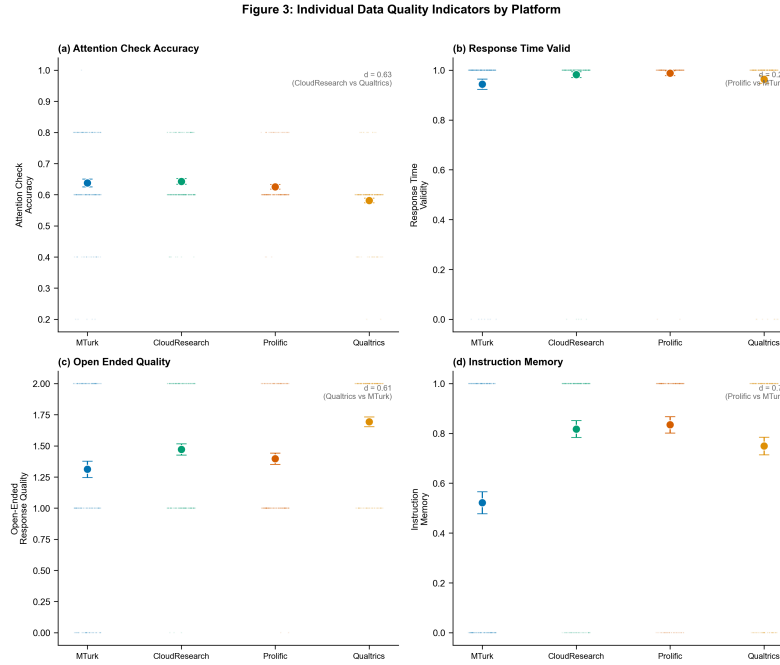


Figure 4: Individual data quality indicators by platform. Panel figure displaying platform means with 95% confidence intervals (error bars) for each of the four primary quality indicators across all 2,076 participants. Semi-transparent points show individual observations (subsampled to 200 per platform for visual clarity). (a) Attention check accuracy: CloudResearch ($M = 0.64$) ranked highest and Qualtrics ($M = 0.58$) lowest, $d = 0.63$. (b) Response time validity: Prolific ($M = 0.99$) ranked highest and MTurk ($M = 0.94$) lowest, $d = 0.24$. (c) Open-ended response quality (scale: 1–2): Qualtrics ($M = 1.69$) ranked highest and MTurk ($M = 1.31$) lowest, $d = 0.61$. (d) Instruction memory: Prolific ($M = 0.83$) ranked highest and MTurk ($M = 0.52$) lowest, $d = 0.71$. The divergence in best- and worst-performing platforms across panels illustrates the multidimensional nature of data quality across online platforms.

2.4 Robustness Checks

Seven robustness checks assessed the sensitivity of the primary findings. First, a Kruskal-Wallis non-parametric test on the post-exclusion composite confirmed the significant platform effect, $H(3) = 17.64$, $p < .001$. Second, as described above, the all-data five-indicator analysis ($N = 2,076$) recovered the classical platform ranking with a large effect. Third, restricting the analysis to participants flagged as high-quality by the original study’s `is_high_quality` filter ($n = 988$) yielded a significant platform effect, $F(3, 984) = 18.29$, $p < .001$. Fourth, an ANCOVA controlling for age, gender, education, and income ($n = 1,107$ after listwise deletion) showed that the platform effect persisted after demographic adjustment, $F(3, 1099) = 7.78$, $p < .001$, $\eta_p^2 = .021$. Fifth, a Bayes Factor computed via the BIC approximation indicated that, in the post-exclusion sample, evidence favored the null model of no platform differences: $BF_{10} = 0.09$, $BF_{01} = 10.54$. This finding—that a significant p -value coexists with Bayesian evidence for the null—reflects the small effect size ($\eta_p^2 = .015$) in a large sample. Sixth, Mann-Whitney U pairwise tests confirmed the CloudResearch-versus-Prolific and CloudResearch-versus-Qualtrics differences ($U = 54,671$, $p < .001$ and $U = 65,608$, $p = .002$, respectively). Seventh, a three-indicator composite excluding open-ended quality (which favors Qualtrics) produced a significant platform effect, $F(3, 1108) = 14.60$, $p < .001$.

Figure 4: Platform Rankings Diverge Across Quality Indicators

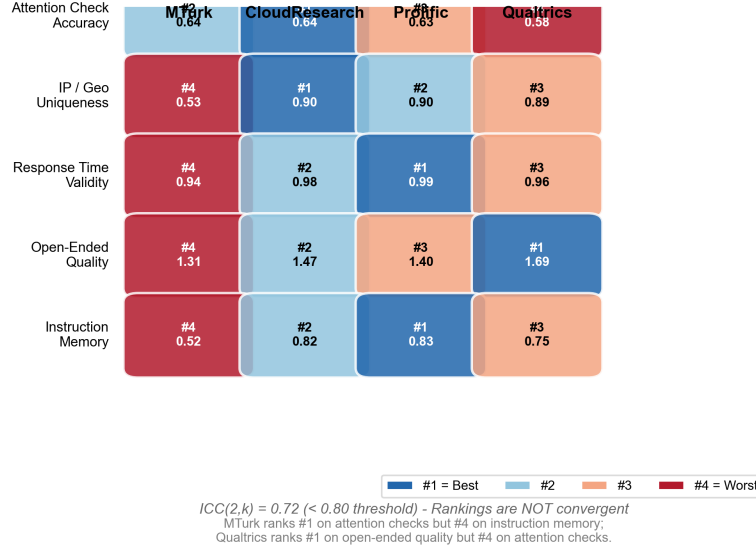


Figure 5: Platform rankings diverge across quality indicators. Heatmap displaying per-platform rankings (1 = best to 4 = worst) for each of five quality indicators computed from all data ($N = 2,076$). Cells are color-coded from blue (best) to red (worst) and annotated with the rank and the raw indicator mean. Rankings reveal substantial divergence: MTurk ranks #1 on attention check accuracy ($M = 0.64$) but #4 on instruction memory ($M = 0.52$), response time validity ($M = 0.94$), and IP/geolocation uniqueness (53.4%). Qualtrics ranks #1 on open-ended response quality ($M = 1.69$) but #4 on attention check accuracy ($M = 0.58$). The intraclass correlation coefficient for average measures was $ICC(2, k) = .72$, below the pre-registered .80 convergence threshold, indicating that the five quality indicators do not measure a unitary construct.

2.5 Untestable Hypotheses

H3 (MTurk quality decline across waves), H4 (Prolific and CloudResearch stability), and H5 (cost-efficiency advantages for research-specific platforms) could not be tested with the available single-wave dataset. Testing these hypotheses requires at least two measurement waves spaced at six-month intervals, with per-participant cost data linked to quality indicators. The analytical framework for these tests—linear mixed-effects models with Platform $\times$ Wave interaction, TOST equivalence testing for stability, and Kruskal-Wallis tests on cost-per-quality-respondent—is specified in the pre-registration and documented in the analysis code.

3 Discussion

We re-analyzed the ? dataset to establish a rigorous baseline for a longitudinal replication tracking data quality across online research platforms. Two findings stand out. First, after applying the original study’s own pre-registered exclusion criteria, the cross-sectional platform ranking that the original paper reported—Prolific and CloudResearch outperforming MTurk and Qualtrics—did not replicate. The planned contrast was not significant ($d = 0.03$, $p = .608$), and the observed rank order placed MTurk second among four platforms, indistinguishable from the top-ranked CloudResearch ($d = -0.05$, $p = .66$). This null result is not evidence that platforms do not differ; rather, it is a direct consequence of a differential selection effect. MTurk participants were excluded at nearly 2.5 times the rate of CloudResearch participants (70.4% vs. 34.3%), truncating the MTurk distribution to its highest-quality tail. When exclusion criteria were suspended in the all-data robustness check, the classical ? ranking re-emerged with a large effect ($\eta_p^2 = .110$). The practical implication is sobering: a researcher who applies the same screening procedures across platforms may, after exclusions, find themselves analyzing data from fundamentally different subpopulations—MTurk’s most diligent workers versus CloudResearch’s broader pool. Platform quality rankings are not a property of platforms alone; they are a joint function of platform features and the exclusion criteria researchers apply.

Second, we found that the five quality indicators used in this literature do not measure a unitary construct. The ICC of .72 fell below the pre-registered convergence threshold of .80, and the pairwise correlations among indicators within participants were effectively zero (all $|r| < .10$). MTurk participants ranked first on attention check accuracy but last on instruction memory, response time validity, and IP/geolocation uniqueness. Qualtrics participants wrote the highest-quality open-ended responses but failed the most attention checks. This indicator divergence has consequences for both methodology and theory. On the methodological side, any composite quality score that assigns equal weight to all indicators will obscure platform-specific failure modes. A researcher using only attention checks to screen participants would rank MTurk favorably; a researcher relying on instruction memory or IP uniqueness would reach the opposite conclusion. The choice of quality indicator is, in effect, a choice about which platform appears best. On the theoretical side, the near-zero inter-correlations among indicators suggest that they capture fundamentally different behavioral phenomena—vigilance to explicit instructions, motivation to write substantive text, capacity to recall previously presented information, and the honesty to provide unique digital identifiers—rather than any single latent dimension of response quality.

These findings must be evaluated in the context of several limitations. The most consequential is the absence of SONA Systems data from the re-analysis. The original study collected SONA data separately at the University of Wisconsin–Madison ($n \approx 500$), and that dataset was not included in the public OSF repository. A full five-platform comparison awaits the release of those data. Second, the dataset represents a single cross-sectional wave collected between May and July 2021—before GPT-3.5 and ChatGPT became publicly available and before several platform policy changes (e.g., MTurk’s transition to requiring AWS billing accounts). Contemporary online samples may differ from this baseline in ways we cannot assess without new data collection. The longitudinal replication design we propose—four waves at six-month intervals—would directly address this unknown. Third, the sample is WEIRD in the original sense (?): all U.S.-based, predominantly White, with above-average education. Platform quality differences observed here may not generalize to non-U.S. worker populations, which differ in economic incentives, language proficiency, and platform availability. Fourth, the study’s quasi-experimental design means that platform membership is confounded with self-selection. Participants who choose MTurk differ from those who choose Prolific on dimensions beyond demographics—motivation, survey experience, trust in research institutions—that could influence data quality independent of platform features. Fifth, we could not test three of our five pre-registered hypotheses (H3–H5) because the available data lack temporal depth and cost information. The analytical infrastructure for these tests is in place, and their resolution awaits future data collection waves.

Despite these limitations, the present re-analysis makes two contributions that bear on research practice. The first is cautionary: the platform ranking a researcher observes is partly under their own control. Applying pre-registered exclusion criteria reshapes the effective sample in platform-specific ways, and what remains after exclusions may not represent the underlying population that the platform originally delivered. This does not mean researchers should abandon screening—the alternative is accepting demonstrably low-quality data—but it does mean that effect sizes computed from retained samples should be interpreted as upper-bound estimates of platform differences, and that the demographic composition of retained samples (rather than recruited samples) should be reported and discussed. The second contribution is methodological: because quality indicators do not converge, researchers should select and pre-register their primary quality indicator based on what matters most for their specific research question. A study of factual knowledge needs instruction memory; a study of attitudes needs open-ended response quality; a study of individual differences needs psychometric consistency. No single indicator, and no equal-weight composite, is adequate for all purposes.

The planned longitudinal replication will extend this baseline in three ways. It will test whether the platform ranking obtained in 2021 still holds in 2026 and beyond (H1). It will determine whether MTurk quality continues its documented decline (H3; ??) while Prolific and CloudResearch remain stable (H4). And it will track whether the cost per high-quality respondent—which ? estimated at \$1.90 for Prolific, \$4.36 for MTurk, and \$8.17 for Qualtrics—has shifted as platform pricing, worker populations, and quality have changed (H5). All data, materials, and analysis code from both the baseline and subsequent waves will be publicly archived on the Open Science Framework. The pre-registration for the full multi-wave study, including all exclusion criteria and analysis plans, will be made available upon registration.